Digital Object Identifier

Joint Channel-and-Architecture Search for EEG Classification of Neurological Disorders

MATEUS R. CRUZ[1], AND TALES C. PIMENTA.[1]
[1]Instituto de Engenharia de Sistemas e Tecnologias da Informação (IESTI), Universidade Federal de Itajuba (Unifei), Itajuba, MG 37500903 BR
Corresponding author: Mateus R. Cruz (e-mail: d2020103671@unifei.edu.br).

This work was supported in part by CAPES, in part by CNPq, and in part by FAPEMIG.

ABSTRACT

Neural Architecture Search (NAS) is a promising strategy for automated model design in electroencephalography (EEG), but clinical adoption remains constrained by limited sample sizes, strong inter-subject variability, and hardware constraints in wearable settings. This work presents a differentiable joint optimization framework that learns EEG channel subsets and neural architectures simultaneously under a multiobjective criterion balancing predictive performance, sensor burden, and structural cost. Experiments were conducted on three public datasets (depression: $N = 119$, $C = 19$; schizophrenia: $N = 84$, $C = 16$; epilepsy: $N = 318$, $C = 14$) using sample-wise stratified 5-fold cross-validation (StratifiedKFold, split seed 42), with normalization fitted on the training partition of each fold only, together with a three-phase training schedule (20/70/20 nominal epochs with phase-wise early stopping). The proposed method achieved consistent channel reduction (68–71%), with reference-run macro-F_1 scores of 0.778 ± 0.075, 0.723 ± 0.089, and 0.602 ± 0.074 for depression, schizophrenia, and epilepsy, respectively. The strongest empirical advantage was observed for schizophrenia, where joint optimization outperformed the reported reference deep baselines under severe data scarcity. For depression and epilepsy, the results indicate an explicit accuracy–efficiency trade-off in which fewer electrodes are obtained at the cost of part of the attainable macro-F_1. The reported ablations further show that removing channel selection increases F_1 but requires substantially more electrodes. Stability analysis across 10 independent runs (seeds 0–9) indicates that channel selection is more consistent than operation selection, with the highest operation consistency observed in the largest cohort. These findings support joint channel-and-architecture optimization as a practical route toward compact, channel-explicit, auditable, and deployable EEG diagnostic systems.

INDEX TERMS EEG, neural architecture search, channel selection, assistive diagnosis, neurological disorders, embedded inference

I. INTRODUCTION

Neural Architecture Search (NAS) has emerged as a major paradigm shift in deep learning by automating the design of neural architectures, consistently surpassing manually engineered models across vision, mobile computing, and natural language processing benchmarks [3]. Representative studies on large-scale image classification and resource-constrained deployment have shown that systematic exploration of the architectural search space can uncover solutions beyond human intuition [2]. In parallel, methodological advances have substantially reduced the computational cost of NAS, decreasing requirements from thousands of GPU-days to levels accessible in academic and clinical research settings [15].

Despite these advances, the adoption of NAS in clinical applications involving biomedical time-series remains limited [14]. Unlike vision or language domains, physiological signals commonly involve small datasets, high inter-subject variability, strongly correlated multi-channel measurements, and demanding interpretability requirements in clinical use [18]. Under these conditions, NAS approaches originally developed for large-scale data and homogeneous architectural spaces become inefficient and poorly aligned with clinical realities.

Electroencephalography (EEG) exemplifies these challenges. As a core modality for diagnosing and monitoring

neurological and psychiatric disorders affecting hundreds of millions of individuals worldwide [4], EEG provides a temporally rich representation of brain dynamics, combining high sampling rates with spatially distributed channels [7]. However, this richness introduces severe combinatorial complexity, as even moderate EEG configurations with 14 channels yield 2^{14} possible channel subsets, which further multiply when combined with heterogeneous architectural paradigms such as convolutional, recurrent, and attention-based models [11].

Beyond combinatorial growth, clinical EEG analysis faces three additional structural barriers. First, the computational cost of traditional NAS is incompatible with clinical small-data regimes, where datasets often include only a few hundred patients [8]. Second, there is no architectural consensus across clinical tasks, with models effective for epilepsy, depression, or schizophrenia differing substantially in structure and inductive bias [5]. Third, clinical deployment requires neurophysiological interpretability, such that selected channels and temporal patterns must correspond to meaningful anatomical brain regions, a requirement largely ignored in conventional NAS pipelines [5].

Early studies applying NAS to EEG reported performance gains in non-clinical domains such as motor imagery and emotion recognition [16]. However, these approaches share critical limitations, including high computational cost, lack of explicit channel selection, restriction to a single architectural search space, and validation on tasks that do not reflect real-world clinical constraints [13]. As a result, existing EEG-NAS methods remain distant from translational clinical use.

In parallel, the emergence of zero-cost proxies has reshaped the NAS landscape by enabling architecture evaluation without full training, relying instead on gradient-based and synaptic-flow measures [9]. Methods based on connection sensitivity and information flow demonstrated strong rank correlation with fully trained performance while reducing evaluation time to milliseconds [10]. Despite their impact, zero-cost NAS methods remain largely unexplored in biomedical time-series, even as recent surveys explicitly call for efficient NAS solutions under clinical resource constraints.

Conversely, EEG channel selection has traditionally been treated as an independent problem, addressed through evolutionary algorithms or classical machine learning techniques [17]. While effective in reducing dimensionality, these methods require full model training for each candidate configuration, rendering joint optimization of architecture and channel selection computationally infeasible [6].

Consequently, a fundamental gap persists: the absence of a unified, computationally efficient, and clinically interpretable framework capable of jointly optimizing neural architectures and EEG channel subsets under severe data and resource constraints [1].

To address this gap, we propose SNIP-it, a differentiable multiobjective NAS framework that integrates architecture search and EEG channel pruning within a single optimization process. By coupling gradient-based channel saliency estimation to explicit sensor, MAC, and parameter penalties during search, the framework jointly learns compact channel masks and heterogeneous neural architectures under a unified clinical NAS paradigm [12], [19].

II. MATERIALS AND METHODS

A. END-TO-END OPERATIONAL ARCHITECTURE

The proposed framework is organized as a closed-loop methodological pipeline for assistive EEG analysis, integrating signal acquisition, compact representation learning, near-real-time inference, data persistence, model updating, and controlled deployment within a single operational cycle. This organization is intended to connect the full trajectory from incoming electrophysiological measurements to model revision, thereby aligning diagnostic support with longitudinal data accumulation and iterative performance improvement.

In the acquisition stage, multichannel EEG is processed through sliding temporal windows of 2 s sampled at 250 Hz, which enables continuous monitoring while preserving temporal locality. Each window then undergoes a feature-extraction stage in which every channel is summarized by five canonical frequency bands (delta, theta, alpha, beta, and gamma) combined with four attributes per band—band power, relative power, spectral entropy, and mean amplitude—yielding a compact representation of size $[C, 20]$. The resulting channel-by-feature tensor is normalized with z-score scaling computed on the training partition of each fold and then applied to the corresponding validation data. This design choice decreases model-inference cost relative to end-to-end raw-signal modeling while preserving clinically relevant spectral information for neurological classification tasks.

The inference stage operates on the compact representation and returns not only the predicted class, but also class probabilities, processing latency, and information about the active electrode configuration used by the model. This expanded output structure supports operational traceability and facilitates downstream auditing of performance, efficiency, and sensor utilization. To ensure service continuity, model access follows a hierarchical loading strategy in which the currently promoted registry version is prioritized while previously validated local artifacts remain available as fallback alternatives, thereby reducing the risk of interruption during model updates.

All processed windows and their associated annotations are stored in a persistent data layer that supports both auditability and future learning. Once labeled samples become available, they are incorporated into an automated updating pipeline that exports curated data, performs retraining and reevaluation, and promotes a new model version only when the target metric improves without violating cost or latency constraints. In this way, the framework implements a controlled form of continual improvement in which model

evolution is conditioned on empirical evidence rather than unconditional replacement.

After a candidate model has been accepted, a deployment stage converts the learned solution into a compact inference-ready form suitable for runtime operation under hardware constraints. The promoted artifact is a slim discrete model accompanied by versioned metadata describing active channels and input configuration. The production environment periodically checks for promoted versions and can replace the active model without interrupting the surrounding service. This end-to-end architecture is particularly relevant for assistive EEG diagnostics because it unifies data collection, efficient prediction, cost-aware deployment, and adaptive updating while preserving the interpretation that all predictions remain supportive rather than definitive clinical decisions.

B. MODEL DEVELOPMENT

This study formulates EEG classification as a joint optimization problem over channel-mask logits θ, backbone weights ϕ, and architectural logits α. The goal is to maximize unweighted macro-F_1 while minimizing sensor burden and structural cost under hardware-constrained inference.

Table 1 summarizes the three public datasets. Each sample is represented as a matrix $[C \times T]$, where $T = 20$ denotes the per-channel feature dimension obtained from five canonical EEG bands (delta, theta, alpha, beta, and gamma) combined with four attributes per band. This compact representation is specifically suited to low-power deployment because the classifier operates on precomputed spectral features rather than on an end-to-end raw-signal model.

TABLE 1. EEG datasets used in this study.

Dataset	N	Channels (C)	Features (T)	Classes	Task
Depression	119	19	20	2	MDD vs. control
Schizophrenia	84	16	20	2	SZ vs. control
Epilepsy	318	14	20	2	Seizure vs. non-seizure

All comparative experiments reported in this manuscript were conducted using sample-wise stratified 5-fold cross-validation (StratifiedKFold) with fixed split seed 42. Normalization statistics were estimated on the training split of each fold and applied to the corresponding validation split only. Consistent with the project validation framework, the main comparative tables report the reference run (seed = 0) as mean±std across the five folds, whereas stability assessment is treated as a separate multi-seed evidence block over 10 independent runs (seeds 0–9) with fold-level control given by `seed + fold_id`. Within each fold, models were trained with mini-batch size 8 using Adam (learning rate 10^{-3}, weight decay 10^{-4}) under a three-phase schedule comprising warm-up, joint search, and discrete fine-tuning. The nominal phase lengths were 20/70/20 epochs, with phase-wise early stopping (patience 15), best-state restoration, and unweighted Macro-F_1 on the concatenated validation fold as the primary selection metric.

TABLE 2. Training protocol.

Setting	Value
Cross-validation	Sample-wise stratified 5-fold (split seed = 42)
Phase schedule	20/70/20 epochs (warm-up/search/fine-tune)
Batch size	8
Optimizer	Adam ($\eta = 10^{-3}$, weight decay $= 10^{-4}$)
MAC coeff.	$\alpha = 0.01$
Sensor/entropy coeff.	$\beta = 0.05$
Param. coeff.	$\lambda_{par} = 0.02$
Min-active penalty	$\gamma = 1.0$, $K = 3$
Adaptive scaling	$\lambda_{eff} = \lambda \min(1, N/1000)$
Early stopping	Patience = 15, phase-wise best-state restore
Arch. LR multiplier	$10\times$
Independent runs	10 (seeds 0–9; fold seed = seed + fold_id)
Default threshold	$\tau = 0.5$
Primary metric	Macro-F_1 on concatenated validation fold

For fixed baselines trained with the standard trainer, the optimization objective is plain cross-entropy. The cost and mask regularizers described below are activated only in the joint-search trainer during phase 2.

To establish robust performance bounds, eleven baselines were evaluated, including fixed architectures, statistical channel selection, differentiable masking without architecture search, NAS with fixed channels, and EEG-specific deep networks (GRU, BiGRU, BiGRU-attention, and ConvNet). The statistical selectors (variance and Fisher) are fit using training data only within each fold, preventing validation leakage during channel ranking. This design isolates the contribution of channel pruning from the contribution of architecture optimization.

Primary cross-dataset performance summaries are reported for the reference run (seed = 0) under the protocol above, whereas dedicated stability analyses follow the framework's multi-seed validation block across 10 independent runs (seeds 0–9).

C. DIFFERENTIABLE JOINT OPTIMIZATION OF CHANNELS AND ARCHITECTURE

The proposed search strategy treats electrode selection and neural architecture design as a single optimization problem, so that predictive performance, sensor burden, and computational efficiency are balanced throughout model development. For input $x \in \mathbb{R}^{C\times T}$ and label y, the operative search-phase objective is

$$\begin{aligned}\min_{\theta,\phi,\alpha}\ & \mathcal{L}_{task}(\theta,\phi,\alpha) + \alpha_{eff}\mathcal{C}_{mac}(\alpha) \\ & + \beta_{eff}\mathcal{C}_{sen}(\theta) + \lambda_{par,eff}\mathcal{C}_{par}(\alpha) \\ & + \mathcal{P}_{min}(\theta) + \beta_{eff}\mathcal{P}_{ent}(\theta)\end{aligned} \tag{1}$$

where θ denotes channel-mask logits, ϕ the backbone weights, and α the architectural logits. This phase-2 objective is the one implemented in the joint-search trainer; phases 1 and 3 revert to $\mathcal{L}_{task}$ only, and the fixed baselines are trained with standard cross-entropy. The formulation allows the representation induced by the retained EEG channels to influence architectural specialization during the same learning

process, producing solutions that are simultaneously compact and task-adaptive.

The channel-selection stage assigns a differentiable relevance weight to each electrode through a sigmoid mask,

$$w_c = \sigma(\theta_c), \qquad \tilde{x}_{b,c,t} = w_c \, x_{b,c,t}, \tag{2}$$

where $\tilde{x}$ is the masked input tensor. Sensor burden is incorporated through a normalized cost term,

$$\mathcal{C}_{sen} = \frac{\sum_{c=1}^{C} w_c s_c}{\sum_{c=1}^{C} s_c}, \tag{3}$$

in which s_c can be uniform, channel-weighted, or region-weighted depending on the acquisition scenario.

Concurrently, the architectural search stage models each computational block as a probabilistic mixture of candidate operations. For cell l, the probability of operation i is

$$p_i^{(l)} = \mathrm{softmax}(\alpha^{(l)})_i, \tag{4}$$

and the default search space includes `conv1d_k3`, `conv1d_k5`, `dsconv1d`, `gru1d`, lightweight attention, and identity mappings. Pooling operators exist in the implementation registry but are not part of the default search space considered in the main configuration. Arithmetic and parameter costs are estimated in expectation as

$$\begin{aligned} \mathcal{C}_{mac} &= \sum_{l=1}^{L} \sum_{i=1}^{|\mathcal{O}|} p_i^{(l)} m_i, \\ \mathcal{C}_{par} &= \sum_{l=1}^{L} \sum_{i=1}^{|\mathcal{O}|} p_i^{(l)} q_i \end{aligned} \tag{5}$$

where m_i and q_i are the precomputed MAC and parameter counts of operation i.

To avoid degenerate sensor configurations, the objective includes two complementary regularizers over the channel mask. In the implementation, the minimum-active penalty acts on the top-K soft channel weights with $K = \texttt{min_active_channels} = 3$ by default,

$$\mathcal{P}_{min} = \gamma \sum_{k \in \text{top-}K(w)} \max(0, 0.5 - k)^2, \tag{6}$$

which discourages collapse toward nearly null channel sets, and

$$\mathcal{P}_{ent} = \frac{1}{C} \sum_{c=1}^{C} \left[-w_c \log(w_c) - (1 - w_c) \log(1 - w_c) \right], \tag{7}$$

which pushes the mask toward polarized values and reduces ambiguity during discretization. In the implemented loss, this entropy term is weighted by the same effective sensor coefficient β_{eff} used for $\mathcal{C}_{sen}$.

Because clinical EEG datasets often operate under severe sample scarcity, the framework uses dataset-size-adaptive regularization coefficients. With default base values $\alpha = 0.01$, $\beta = 0.05$, $\lambda_{par} = 0.02$, and $N_{ref} = 1000$, the effective weights are computed as

$$\lambda_{eff} = \lambda \cdot \min\left(1, \frac{N}{N_{ref}}\right), \qquad N_{ref} = 1000, \tag{8}$$

so that smaller cohorts receive milder penalization than larger ones. Adaptive scaling is enabled by default for α, β, and λ_{par}, whereas the quadratic minimum-active penalty keeps $\gamma = 1.0$ fixed.

After convergence, the continuous solution is transformed into a deployable discrete model by retaining channel c when $\sigma(\theta_c) > 0.5$ and selecting, in each cell, the operation with maximum architectural logit. The resulting model therefore combines a reduced electrode set with an architecture specialized for the target disorder, yielding a clinically oriented compromise among diagnostic discrimination, acquisition practicality, and inference efficiency. Consequently, the final number of active channels is not fixed a priori; it emerges from the learned logits at the default discretization threshold $\tau = 0.5$.

D. THREE-PHASE TRAINING PROTOCOL

To stabilize joint optimization in low-sample EEG scenarios, the training procedure is organized into three successive phases that separate representation consolidation, structural adaptation, and final calibration. This staged design reduces interference between predictive learning and compactness constraints, while improving consistency between the continuous search process and the discrete model used during deployment.

The first phase is devoted to representation warm-up and uses a nominal budget of 20 epochs. During this stage, the predictive pathway is refined while architectural parameters remain frozen and no MAC, parameter, or sensor penalties are applied, i.e., $\alpha_{eff} = \beta_{eff} = \lambda_{par,eff} = 0$. The effective objective is therefore

$$\mathcal{L}^{(1)} = \mathcal{L}_{task}, \tag{9}$$

which allows the backbone to learn a physiologically informative representation before compression pressures are introduced.

The second phase performs the joint search itself and uses a nominal budget of 70 epochs. In this stage, backbone and search parameters are jointly optimized with two optimizers, and the effective loss per batch becomes

$$\begin{aligned} \mathcal{L}^{(2)} = \mathcal{L}_{task} &+ \alpha_{eff} \mathcal{C}_{mac} + \beta_{eff} \mathcal{C}_{sen} \\ &+ \lambda_{par,eff} \mathcal{C}_{par} + \mathcal{P}_{min} + \beta_{eff} \mathcal{P}_{ent} \end{aligned} \tag{10}$$

By default, the implementation uses $\alpha = 0.01$, $\beta = 0.05$, $\lambda_{par} = 0.02$, $\gamma = 1.0$, $\texttt{min_active_channels} = 3$, and an architectural learning-rate multiplier of $10\times$. Compactness-oriented regularization may be delayed until a

target fraction of the desired validation F_1 is reached, which helps preserve stability in severely data-constrained regimes.

The third phase is dedicated to discrete fine-tuning and uses a nominal budget of 20 epochs. After completion of the continuous search, the model activates a discrete forward path, freezes architectural parameters, and retrains only the backbone with the discretized channel mask and selected operations. As in phase 1, $\alpha_{eff} = \beta_{eff} = \lambda_{par,eff} = 0$, so the effective objective returns to

$$\mathcal{L}^{(3)} = \mathcal{L}_{task}. \tag{11}$$

Across all phases, progression is governed by validation Macro-F_1 with phase-wise early stopping (patience 15), checkpointing of the best phase state, and restoration of the corresponding weights before the next stage begins. This protocol limits overfitting, reduces unnecessary computation, and narrows the gap between the relaxed supernet and the final exportable model.

E. DISEASE-SPECIFIC MULTIOBJECTIVE MODEL SELECTION

The identification of the most appropriate architecture for each neurological disorder is formulated as a multiobjective selection procedure rather than a single-metric comparison. Candidate models are assessed jointly with respect to diagnostic discrimination, electrode economy, computational burden, temporal efficiency, and reproducibility across repeated evaluations. This formulation is motivated by the fact that clinically usable EEG systems must preserve predictive accuracy while remaining compatible with reduced montages and resource-constrained deployment.

Disease-specific model selection begins by acknowledging heterogeneity among cohorts, including differences in the number of available channels, class balance, and sample size. These factors influence the level of representational complexity that can be supported by the data and therefore require adaptive capacity assessment during model comparison. When capacity search is enabled, the framework evaluates hidden dimensions in $\{32, 48, 64\}$ and numbers of search cells in $\{3, 4\}$, allowing the final recommendation to reflect both signal characteristics and data availability.

The decision process follows a constrained hierarchical strategy centered on predictive performance and deployment cost. In the implementation used here, candidate solutions are compared lexicographically by Macro-F_1, sensor burden, expected MACs and parameter count, stability across runs, and latency. When required for auditability, the same criteria may also be summarized through a normalized scalar score,

$$S = w_1\hat{F}1 - w_2\hat{C}_{sen} - w_3\hat{C}_{mac} - w_4\hat{C}_{lat} + w_5\hat{E}_{stab}, \tag{12}$$

where hatted quantities are normalized to $[0, 1]$. The scalar score is auxiliary; the main decision remains grounded in hierarchical multi-criteria comparison.

To support the final recommendation for each disorder, candidate models are further subjected to complementary evidence procedures. These evidence blocks include component-removal analysis, robustness under corruption, reproducibility across repeated runs, and sensitivity to mask discretization. At the model level, channel selection is not defined by a fixed pre-specified K; instead, the effective number of active channels emerges after training as the number of entries satisfying $\sigma(\theta_c) > 0.5$ at discretization, while each search cell selects the operation with the maximum architectural logit. This designation should be interpreted as the best computational configuration within the experimental protocol, rather than as definitive clinical validation.

F. EXPERIMENTAL VALIDATION AND ROBUSTNESS ASSESSMENT

The methodological validation strategy is designed to demonstrate not only predictive effectiveness under ideal conditions, but also reliable behavior under realistic variations in EEG acquisition. In accordance with the project documentation, model assessment combines sample-wise stratified partitioning via StratifiedKFold for the main comparative tables, dedicated multi-seed runs for stability analysis, and a primary emphasis on unweighted Macro-F_1. The current implementation does not use GroupKFold or subject-level grouping, so the reported estimates should be interpreted as sample-level stratified validation results.

Comparative evaluation is performed against complementary reference groups, including fixed architectures, models coupled with non-adaptive channel reduction, sequential optimization pipelines, and the proposed joint-search approach. Statistical channel-selection baselines such as variance and Fisher ranking are computed using only the training subset within each fold. This comparative structure makes it possible to distinguish gains attributable to joint structural adaptation from gains arising solely from feature reduction or model capacity. As a result, the experimental protocol supports a clearer interpretation of the contribution of each methodological component.

The broader validation framework is organized around four complementary evidence blocks: ablation, robustness under corruption, stability across repeated runs, and sensitivity to mask discretization. In the present manuscript, the reported evidence blocks cover ablation, stability, and fixed-threshold sensitivity ($\tau = 0.3, 0.5, 0.7$). The codebase also supports top-k discretization sweeps, but those results are not reported here.

Statistical uncertainty in the reported tables is summarized through mean and standard deviation. Operationally, Macro-F_1 is computed once per fold over the concatenated validation logits and targets, averaging classwise F1 over the classes present in that fold. For a given run, the reported value is then mean$\pm$std across the five folds; for cross-seed summaries, the reported value is the mean of the per-seed fold means across the 10 independent runs. Matched comparisons by seed/fold are used whenever a direct contrast between can-

didate methods is required, thus preserving correspondence between evaluation conditions. This reporting strategy improves interpretability and prevents isolated high-performing runs from being mistaken as representative behavior.

Within the present report, a configuration is considered methodologically stronger when it delivers higher fold-level Macro-F_1, preserves a non-degenerate channel subset after discretization, and remains stable across repeated runs relative to the analytic random-reference Jaccard. When corruption tests are performed, predictable rather than erratic degradation is treated as additional evidence of robustness. Taken together, these criteria ensure that the preferred model is supported by accuracy, reproducibility, and resilience, which are essential properties for translation from controlled experimentation to assistive EEG applications.

III. RESULTS AND DISCUSSION

A. BASELINE PERFORMANCE ACROSS DATASETS

Table 3 reports representative baseline results. Two consistent trends emerge. First, recurrent models are strongly competitive in depression and epilepsy, where GRU and BiGRU reach $F_1 = 0.880 \pm 0.037$ and 0.839 ± 0.022, respectively. Second, schizophrenia remains the hardest regime ($N = 84$), in which the best baseline is a statistical selector (variance-based, $F_1 = 0.621 \pm 0.102$), indicating that high-capacity temporal models are more sensitive to fold-level sampling variance under severe data scarcity.

TABLE 3. Cross-dataset baseline comparison (reference run seed = 0; mean±std across stratified folds).

Method	Depression	Schizophrenia	Epilepsy	Channels
All channels (fixed arch.)	0.544 ± 0.169	0.556 ± 0.218	0.713 ± 0.133	100%
Variance selection	0.702 ± 0.283	0.621 ± 0.102	0.769 ± 0.045	50%
NAS (fixed channels)	0.762 ± 0.078	0.607 ± 0.125	0.787 ± 0.084	100%
Best EEG-DL per dataset	0.880 ± 0.037	0.464 ± 0.241	0.839 ± 0.022	100%
Joint search (ours)	0.778 ± 0.075	$\mathbf{0.723 \pm 0.089}$	0.602 ± 0.074	29–36%

Model-size analysis reinforces a complexity–generalization trade-off. In depression, a compact GRU (20,738 parameters) outperformed larger BiGRU and attention-enhanced variants (41,282 and 45,442 parameters), indicating that increased capacity can amplify variance when the per-channel feature axis is compact ($T = 20$) and N is limited. In epilepsy, where $N = 318$, larger recurrent capacity became beneficial, consistent with the lower fold-to-fold variability observed in Table 3.

The differentiable-mask-only baseline exhibited a recurrent collapse to 0–2 active channels ($F_1 \approx 0.33$–0.40 across datasets), confirming that standalone ℓ_1-dominated mask optimization is unstable in low-data settings and motivating the regularized joint formulation.

B. JOINT CHANNEL-AND-ARCHITECTURE OPTIMIZATION

The core joint-search outcomes are summarized in Table 4. Across all disorders, the model selected only 4–6 channels, corresponding to a 68–71% reduction in electrodes. This reduction is clinically relevant because electrode count directly affects setup time, gel usage, and patient comfort during ambulatory acquisition.

Unless otherwise stated, the values in Tables 4 and 5 are fold-level mean±std from the reference run (seed = 0). Table 7 is obtained by post-hoc discretization of the same trained continuous model without retraining, whereas Table 8 summarizes independent ablation executions.

TABLE 4. Joint-search performance and channel reduction (reference run seed = 0; mean±std across stratified folds, with best-fold F_1 reported separately).

Dataset	Joint F_1	Best-fold F_1	Channels	Reduction
Depression	0.778 ± 0.075	0.873	6/19	68%
Schizophrenia	0.723 ± 0.089	0.821	5/16	69%
Epilepsy	0.602 ± 0.074	0.752	4/14	71%

To contextualize these results, Table 5 compares the joint model with a sequential channel-then-architecture baseline under the same sample-level stratified protocol used throughout. Because the implemented validation split does not enforce subject grouping, these values should be interpreted as sample-level comparative estimates rather than as subject-level generalization results. Although the sequential approach is numerically higher in all datasets, the depression score (1.000 ± 0.000) under 5-fold validation with approximately 24 validation samples per fold should still be interpreted cautiously and verified across repeated runs. The joint objective, while less optimistic numerically, behaves as an implicit regularizer and avoids similarly degenerate operating points in the reference run.

TABLE 5. Joint versus sequential optimization (reference run seed = 0; mean±std across stratified folds).

Dataset	Joint F_1	Sequential F_1	Δ (Joint–Seq.)
Depression	0.778 ± 0.075	1.000 ± 0.000	-0.222
Schizophrenia	0.723 ± 0.089	0.818 ± 0.113	-0.095
Epilepsy	0.602 ± 0.074	0.737 ± 0.055	-0.135

The epilepsy gap ($\Delta = -0.135$) is more informative than the depression gap and suggests a genuine optimization burden: jointly optimizing architecture and channel masks creates a harder loss landscape than two-stage pipelines when model capacity is high.

C. STABILITY AND THRESHOLD SENSITIVITY

Table 6 reports the stability block of the validation framework, summarized here by mean Jaccard overlap between channel masks and operation consistency per search cell across 10 independent runs (seeds 0–9). The random-reference Jaccard is computed analytically as $J = \frac{p^2}{2p-p^2}$ with $p = k/C$, where k is the mean number of active channels observed across runs for the corresponding dataset. Mean Jaccard values are above the reported random-reference values, suggesting that channel importance is not

random across seeds. In contrast, operation consistency is dataset-dependent and is highest in epilepsy.

TABLE 6. Stability analysis across 10 independent runs (seeds 0–9).

Dataset	Mean Jaccard	Random Jaccard	Cell-op consistency
Depression	0.320	∼0.16	0.000
Schizophrenia	0.289	∼0.16	0.000
Epilepsy	0.294	∼0.16	0.250

This dissociation between stable channel masks and unstable architecture choices indicates that spatially informative electrodes can be identified under low N, whereas architecture ranking is less consistent in the smaller cohorts studied here. In the present study, the most consistent architecture-level signal was observed in the largest cohort ($N = 318$), but broader validation across additional datasets is still needed.

Table 7 reports the fixed-threshold portion of the discretization-sensitivity analysis. This block is a post-processing analysis of the same trained continuous solution: no retraining is performed, and only the inference-time discretization threshold is changed, with $\tau \in \{0.3, 0.5, 0.7\}$ and default deployment threshold $\tau = 0.50$. These values are therefore not the same statistic as the fold-level mean±std reported in Table 4; instead, they are post-hoc single-model outcomes used to diagnose threshold dependence. The codebase also supports top-k discretization sweeps, but only the fixed-threshold analysis is reported here. The default deployment threshold is also the best setting for depression and epilepsy in these runs. For schizophrenia, $\tau = 0.30$ (retaining all channels) yields the highest unconstrained F_1; however, that operating point removes the sensor-efficiency benefit, so the reported 5/16-channel result in Table 4 reflects the default multiobjective deployment rule at $\tau = 0.50$ rather than the purely F_1-maximizing threshold.

TABLE 7. Threshold sensitivity under post-hoc mask discretization (single-model F_1 without retraining, with number of selected channels in parentheses).

Dataset	$\tau = 0.30$	$\tau = 0.50$	$\tau = 0.70$	Best τ
Depression	0.333 (19)	0.873 (6)	0.333 (0)	0.50
Schizophrenia	0.764 (16)	0.702 (5)	0.346 (0)	0.30
Epilepsy	0.425 (14)	0.538 (4)	0.360 (0)	0.50

The systematic failure at $\tau = 0.70$ (all channels removed) indicates that learned mask probabilities remain mostly in the mid-sigmoid range under regularization.

D. ABLATION STUDY AND TRADE-OFF INTERPRETATION

Table 8 isolates each component contribution. The active-channel counts reported in this table correspond to the mean discretized counts within the ablation runs and therefore should not be interpreted as the final deployment montage summarized in Table 4. The most consistent observation is that removing channel selection yields the highest F_1 in both datasets reported, but only by restoring all electrodes.

Unlike Table 7, these ablation values come from independent executions rather than from post-hoc rediscretization of the reference run.

TABLE 8. Ablation study under the same stratified 5-fold protocol (independent executions).

Dataset	Configuration	Mean F_1	Active channels
Depression	full_model	0.621±0.263	3
Depression	no_sensor_cost	0.582±0.231	12
Depression	no_hw_cost	0.685±0.207	4
Depression	no_channel_selection	**0.958±0.029**	19
Depression	no_nas	0.687±0.219	4
Depression	reduced_search	0.612±0.240	4
Epilepsy	full_model	0.586±0.075	2
Epilepsy	no_sensor_cost	0.685±0.053	7
Epilepsy	no_hw_cost	0.533±0.118	3
Epilepsy	no_channel_selection	**0.898±0.043**	14
Epilepsy	no_nas	0.633±0.046	2
Epilepsy	reduced_search	0.539±0.057	3

These ablations support two complementary interpretations. First, mask and NAS gradients can interfere during joint training, reducing pure predictive accuracy. Second, the methodological target is not unconstrained F_1 maximization; instead, it is Pareto-efficient operation under acquisition constraints. From this perspective, sacrificing part of the maximum achievable F_1 to obtain a substantially smaller electrode set is an intended design outcome rather than a failure mode.

E. EMBEDDED DEPLOYMENT PROFILE

Table 9 reports deployment-oriented characteristics of the resulting models. Latency was measured as CPU-only wall-clock forward-pass time with batch size 1, 10 warm-up iterations, and 50 timed runs using `time.perf_counter()`. The reported values include only the PyTorch forward pass on the $[B, C, T]$ tensor and therefore exclude feature extraction, data loading, and other preprocessing. Because the code does not fix CPU thread count or pin a specific processor, these timings should be read as comparative measurements rather than device-invariant guarantees. All variants remain near 260k parameters and satisfy the low-memory deployment target emphasized by the framework ($\leq$2 MB flash). The reported FP32 inference latency ranges from 4.46 to 7.02 ms.

TABLE 9. Embedded deployment characteristics of joint-search models.

Dataset	Params	FP32 (KB)	INT8 (KB)	FP32 latency (ms)	Char
Depression	∼260k	1073	∼266	7.02	6/
Schizophrenia	∼260k	1073	∼266	4.48	5/
Epilepsy	∼260k	1072	∼266	4.46	4/

The current dynamic-quantization pipeline did not fully exploit theoretical compression limits. Post-training static quantization with calibration data is therefore a practical next step to improve memory reduction without retraining.

F. SYNTHESIS AND LIMITATIONS

Overall, the proposed framework delivers a robust sensor-efficiency gain with clinically meaningful accuracy, with the clearest empirical advantage in schizophrenia relative to the reported deep baselines under low N. The central contribution is therefore a controllable performance–efficiency frontier, not absolute F_1 maximization.

The primary limitations are that the main cross-dataset performance comparison is still anchored to a reference run and that the reported cross-validation is sample-stratified rather than GroupKFold by subject. Ongoing broader multi-seed experiments and subject-grouped evaluation are required to replace fold-based uncertainty with stronger run-to-run and subject-level generalization estimates. In addition, operation consistency remains limited in the two smallest datasets, indicating that future work should combine reduced search spaces with bilevel optimization and stronger priors for low-sample regimes.

G. CRITICAL CONSISTENCY ASSESSMENT

The new results both reinforce and challenge the initial hypotheses, and this dual outcome is scientifically informative rather than contradictory. First, the findings reinforce the main translational claim: as shown in Table 4, the method consistently achieves large channel reduction (68–71%) while maintaining usable diagnostic performance, and Table 9 reports memory and latency values aligned with the low-burden deployment objective. These results support the practical objective of low-burden EEG acquisition.

Second, the results challenge any strong claim that joint optimization universally maximizes predictive accuracy. Table 8 shows that, in the reported ablations, removing channel selection (i.e., using all electrodes) yields the highest F_1 in depression and epilepsy, and Table 5 shows higher sequential scores across datasets. However, this apparent contradiction is resolved by the optimization target itself: the proposed formulation is multiobjective and intentionally penalizes sensor and hardware cost. Therefore, lower unconstrained F_1 relative to all-channel configurations is expected at the Pareto-efficient operating point.

Third, operation-level stability partially challenges the notion of uniformly reliable architecture discovery in small cohorts. Table 6 indicates that channel-selection stability is consistently above random, whereas operation consistency remains low in depression and schizophrenia. This suggests that, under small N, the framework is more reliable for identifying informative spatial subsets than for achieving consistent operation selection.

Taken together, these results refine the paper's interpretation: the framework should be viewed as a principled trade-off mechanism for sensor-efficient deployment, with strongest evidence for channel prioritization and more limited evidence for architecture convergence in the reported datasets.

IV. CONCLUSION

This work presented a joint channel-and-architecture optimization framework for EEG classification in depression, schizophrenia, and epilepsy under realistic clinical data constraints. Across all datasets, the method consistently reduced electrode usage by 68–71%, reaching compact configurations of only 4–6 channels while preserving competitive macro-F_1.

The empirical analysis demonstrates that the approach is particularly effective in the schizophrenia setting, where it outperformed reference deep baselines despite severe sample scarcity. For depression and epilepsy, the results explicitly reveal a controlled contradiction: maximum predictive accuracy is obtained when all channels are retained, whereas joint optimization improves acquisition efficiency at the cost of part of the attainable F_1. This trade-off is consistent with the multi-objective formulation rather than a methodological failure.

Stability and ablation analyses indicate that channel selection is more consistent than architecture selection in the reported runs. These findings suggest a practical guideline for translational use: in small cohorts, joint search is most reliable as a channel-prioritization mechanism, while larger cohorts better support full architecture adaptation. Overall, the proposed framework advances EEG-NAS toward more compact, interpretable, and hardware-efficient diagnostic systems.

REFERENCES

[1] Ali Mohammad Alqudah and Zahra Moussavi. A review of deep learning for biomedical signals: Current applications, advancements, future prospects, interpretation, and challenges. Computers, Materials & Continua, 83(3), 2025.

[2] Chunlei Chen, Peng Zhang, Huixiang Zhang, Jiangyan Dai, Yugen Yi, Huihui Zhang, and Yonghui Zhang. Deep learning on computational-resource-limited platforms: A survey. Mobile Information Systems, 2020(1):8454327, 2020.

[3] Xuanyi Dong, David Jacob Kedziora, Katarzyna Musial, and Bogdan Gabrys. Automated deep learning: Neural architecture search is not the end. Foundations and Trends in Machine Learning, 17(5):767–920, 2024.

[4] Evgenia Gkintoni, Apostolos Vantarakis, and Philippos Gourzis. Neuroimaging insights into the public health burden of neuropsychiatric disorders: a systematic review of electroencephalography-based cognitive biomarkers. Medicina, 61(6):1003, 2025.

[5] Harald Hampel, Peng Gao, Jeffrey Cummings, Nicola Toschi, Paul M Thompson, Yan Hu, Min Cho, and Andrea Vergallo. The foundation and architecture of precision medicine in neurology and psychiatry. Trends in Neurosciences, 46(3):176–198, 2023.

[6] Dongyoon Han, Sangdoo Yun, Byeongho Heo, and YoungJoon Yoo. Rethinking channel dimensions for efficient model design. In Proceedings of the IEEE/CVF conference on Computer Vision and Pattern Recognition, pages 732–741, 2021.

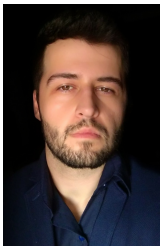
[7] Bin He, Abbas Sohrabpour, Emery Brown, and Zhongming Liu. Electrophysiological source imaging: a noninvasive window to brain dynamics. Annual review of biomedical engineering, 20(1):171–196, 2018.

[8] Georgia Koppe, Andreas Meyer-Lindenberg, and Daniel Durstewitz. Deep learning for small and big data in psychiatry. Neuropsychopharmacology, 46(1):176–190, 2021.

[9] Arjun Krishnakumar, Colin White, Arber Zela, Renbo Tu, Mahmoud Safari, and Frank Hutter. Nas-bench-suite-zero: Accelerating research on zero cost proxies. Advances in Neural Information Processing Systems, 35:28037–28051, 2022.

[10] Sreejan Kumar, Kwangsun Yoo, Monica D Rosenberg, Dustin Scheinost, R Todd Constable, Sheng Zhang, Chiang-Shan R Li, and Marvin M Chun. An information network flow approach for measuring functional

connectivity and predicting behavior. Brain and behavior, 9(8):e01346, 2019.
[11] Wei Li, Penglu Zhao, Cheng Xu, Yingting Hou, Wenhao Jiang, and Aiguo Song. Deep learning for EEG-based visual classification and reconstruction: Panorama, trends, challenges and opportunities. IEEE Transactions on Biomedical Engineering, 2025.
[12] Hou-I Liu, Marco Galindo, Hongxia Xie, Lai-Kuan Wong, Hong-Han Shuai, Yung-Hui Li, and Wen-Huang Cheng. Lightweight deep learning for resource-constrained environments: A survey. ACM Computing Surveys, 56(10):1–42, 2024.
[13] Xin Liu, Jie Li, Jianwei Zhao, Bin Cao, Rongge Yan, and Zhihan Lyu. Evolutionary neural architecture search and its applications in healthcare. CMES-Computer Modeling in Engineering & Sciences, 139(1):143–185, 2024.
[14] Sweet Naskar, Suraj Sharma, Ketousetuo Kuotsu, Suman Halder, Goutam Pal, Subhankar Saha, Shubhadeep Mondal, Ujjwal Kumar Biswas, Mayukh Jana, and Sunirmal Bhattacharjee. The biomedical applications of artificial intelligence: an overview of decades of research. Journal of Drug Targeting, 33(5):717–748, 2025.
[15] Sasan Salmani Pour Avval, Nathan D Eskue, Roger M Groves, and Vahid Yaghoubi. Systematic review on neural architecture search. Artificial Intelligence Review, 58(3):73, 2025.
[16] Ivan Velasco, A Sipols, C Simon De Blas, Luis Pastor, and Sofia Bayona. Motor imagery EEG signal classification with a multivariate time series approach. BioMedical Engineering OnLine, 22(1):29, 2023.
[17] Jianhua Yang, Harsimrat Singh, Evor L Hines, Friederike Schlaghecken, Daciana D Iliescu, Mark S Leeson, and Nigel G Stocks. Channel selection and classification of electroencephalogram signals: an artificial neural network and genetic algorithm-based approach. Artificial intelligence in medicine, 55(2):117–126, 2012.
[18] Xinyang Zhang, Haimin Zhang, and Min Xu. Multimodal classification algorithms for emotional stress analysis with an ECG-centered framework: A comprehensive review. AI, 7(2):63, 2026.
[19] Yu Zhang and Zhe Sage Chen. Harnessing electroencephalography connectomes for cognitive and clinical neuroscience. Nature Biomedical Engineering, 9(8):1186–1201, 2025.



MATEUS CRUZ Mateus Cruz received a M.S. in Telecommunications from Instituto Nacional de Telecomunicações (INATEL) in 2020. He is currently a Ph.D. candidate in the Department of Microelectronics at Universidade Federal de Itajubá (UNIFEI). His research interests include ML, data mining, and artificial intelligence.



TALES PIMENTA TALES C. PIMENTA (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees in electrical engineering from the Federal University of Itajubá, in 1985 and 1988, respectively, and the Ph.D. degree in electrical and computer engineering from Ohio University, in 1992. He held the first visiting scholar position in low-voltage analog integrated circuits with The Ohio State University, in 1997, the second position in the area of ultra-high frequency integrated circuits with Virginia Polytechnic Institute and State University, in 2005, and the third position in the area of devices for biomedical applications with North Florida University. He is currently a Professor with the Federal University of Itajubá. His research interests include circuits and systems for biomedical applications.